Code No: 135SC **RA**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech III Year I Semester Examinations, August/September - 2024
OPERATING SYSTEMS
(Computer Science and Engineering)

Time: 3 hours **Max. Marks: 75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

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PART – A

(25 Marks)

1.a)	Write any two operating system services.	[2]
b)	Distinguish between batch systems and time sharing systems.	[3]
c)	Explain Process Control Block (PCB).	[2]
d)	What is the sequence of operations by which a process utilizes a resource?	[3]
e)	What is meant by Memory Compaction?	[2]
f)	Explain how segment table is implemented?	[3]
g)	List various layers of a file system.	[2]
h)	What are the allocation methods of a disk space?	[3]
i)	How Resource Trajectories can be helpful in avoiding the deadlock?	[2]
j)	What is meant by an Access control list?	[3]

PART – B

(50 Marks)

2.a) What are the functions of Operating Systems? Brief about each.
b) Explain briefly User and Operating System Interface. [5+5]

OR

3.a) What are the advantages of layered structure over monolithic structure?
b) Distinguish between the client-server and peer-to-peer models of distributed system. [5+5]

4.a) What is the need for Inter Process Communication? List the different IPC mechanisms.
b) Define CSP and explain Peterson solution for it. [5+5]

OR

5. What is a monitor? Give the implementation of Bounded Buffer Producer Consumer problem using Monitor. [10]

6. Write short note on:
a) Segmentation
b) Page table
c) Compaction
d) Fragmentation. [2+3+2+3]

OR

7. Explain following page replacement algorithm in detail.
   a) LRU
   b) FIFO. [5+5]

8.a) What is free space management technique?
   b) Explain the implementation of directory using Hash table. [5+5]

OR

9. Explain various disk scheduling algorithms giving their pros and cons. [10]

10. Illustrate Dead lock Avoidance (Banker's Algorithm) with example. [10]

OR

11.a) Explain Protection Mechanism by the use of Protection Domain and Access Control List.
   b) Compare access control list against access matrix. [5+5]

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